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Pinellia ternata lectin induces mitochondrial damage and inflammation in RAW264.7 cells via mannose receptor-mediated endocytosis

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ABSTRACT

A lectin from *Pinellia ternata* tubers (PTL, 12 kDa) with potent pro-inflammatory activity binds to the plasma membrane and translocates into the cytoplasm. However, its cellular entry route and inflammatory toxicity mechanisms remain unclear. Here, confocal microscopy showed that fluorescein isothiocyanate (FITC)-labeled PTL (50 μg/mL) accumulated on the cell surface and was endocytosed by RAW264.7 cells time-dependently. Pharmacological inhibition revealed that PTL uptake was mediated by the mannose receptor C-type 2 (MRC2) and predominantly via clathrin-mediated endocytosis. After endocytosis, PTL translocated to mitochondria with marked co-localization, inducing mitochondrial damage characterized by membrane potential dissipation, calcium overload, and excessive mitochondrial reactive oxygen species (mtROS) generation. These alterations promoted intracellular ROS accumulation and amplified inflammatory responses. Co-immunoprecipitation with mass spectrometry identified ATP synthase subunit beta (ATP5B) as a mitochondrial interacting protein of PTL. Their interaction was validated by molecular docking, co-localization, and microscale thermophoresis (MST). ATP5B knockdown markedly attenuated PTL (50 μg/mL)-induced ROS production, inflammatory cytokine expression, and mitochondrial injury, indicating ATP5B is a critical mediator of PTL's inflammatory toxicity. Collectively, these findings demonstrate that PTL induces mitochondrial damage and pro-inflammatory responses in RAW264.7 cells through MRC2-mediated, clathrin-dependent endocytosis and subsequent ATP5B interaction. This study establishes a mechanistic link between lectin endocytosis, mitochondrial targeting, and inflammatory toxicity, providing new insight into plant lectin-mediated immunotoxicity.

1. Introduction

Pinellia ternata (Thunb.) Breit is a traditional Chinese medicinal herb widely used for the treatment of disorders associated with dampness, phlegm retention, nausea, vomiting, and nodular swelling (Bai et al., 2022). Processed products of *Pinellia ternata* are commonly applied in clinical practice and exhibit considerable therapeutic efficacy. However, the raw tuber contains substantial amounts of *Pinellia ternata* lectin (PTL), a lectin protein considered to be one of the principal toxic constituents responsible for the irritating inflammatory reactions induced by raw *Pinellia ternata*, thereby greatly limiting its clinical application (Wang et al., 2019).

PTL belongs to the lectin family, a diverse group of carbohydrate-binding proteins characterized by their ability to specifically recognize and bind glycoconjugates. Lectins are involved in a variety of important biological processes, including cell recognition, intercellular communication, immune regulation, and signal transduction (Liu et al., 2023). In addition, numerous plant lectins exhibit a broad spectrum of biological

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activities, such as antiviral, antifungal, insecticidal, and antitumor effects (de Oliveira Dos Santos et al., 2022; Duan et al., 2018; Lagarda-Diaz et al., 2017; Sinha et al., 2017). Despite these beneficial activities, accumulating evidence has demonstrated that certain plant lectins also possess pronounced cytotoxic and pro-inflammatory properties, particularly following cellular endocytosis (Landi et al., 2021; Wang et al., 2019). Among these, ricin is one of the most extensively studied examples, exerting potent cytotoxicity through receptor-mediated endocytosis followed by ribosomal RNA depurination, ultimately leading to inhibition of protein synthesis and cell death (Sowa-Rogozinska et al., 2019). These findings suggest that endocytic uptake is a critical determinant of lectin-induced intracellular toxicity.

Previous studies demonstrated that PTL significantly activated RAW264.7 macrophages in a dose-dependent manner, promoted intracellular reactive oxygen species (ROS) accumulation, and triggered activation of the nuclear factor kappa B (NF-κB) signaling pathway, thereby inducing the release of pro-inflammatory cytokines, including TNF-α and IL-1β (Yu et al., 2015). However, despite increasing evidence supporting the pro-inflammatory activity of PTL, the precise cellular entry pathway and the molecular mechanisms underlying PTL-induced inflammatory toxicity remain largely unclear.

Endocytosis is a highly regulated cellular process responsible for the uptake and intracellular trafficking of extracellular macromolecules. Among the various endocytic pathways, receptor-mediated endocytosis plays a crucial role in determining the intracellular fate and biological effects of exogenous ligands. The mannose receptor, also known as mannose receptor C-type 1 (MRC1/CD206), is a member of the C-type lectin receptor family predominantly expressed on macrophages and dendritic cells. This receptor specifically recognizes ligands containing terminal mannose, fucose, or *N*-acetylglucosamine residues and subsequently mediates ligand endocytosis through clathrin-dependent endocytosis (Stavenhagen et al., 2022). In addition to its role in pathogen recognition and antigen uptake, the mannose receptor has also been implicated in the regulation of oxidative stress and inflammatory signaling (Shepherd and Hoidal, 1990; Van Dyken and Locksley, 2013). Previous studies reported that MRC2 expression was significantly upregulated in RAW264.7 macrophages following PTL stimulation (Wang, 2019), suggesting that MRC2 may participate in PTL recognition and intracellular transport.

Notably, the mannose receptor family comprises multiple members, including MRC1 (CD206) and MRC2 (also known as CD280 or Endo180) (East and Isacke, 2002). Although both MRC1 and MRC2 are C-type lectin receptors that mediate endocytosis via the clathrin-dependent pathway, they differ markedly in their ligand specificity and biological functions. MRC1 primarily recognizes mannosylated ligands on pathogens and mediates immune responses (Van der Zande et al., 2021), whereas MRC2 plays a key role in extracellular matrix remodeling by mediating the internalization and lysosomal degradation of collagen (Melander et al., 2015). Given that PTL was a mannose-binding lectin, we hypothesized that MRC2, in addition to MRC1, may serve as a potential receptor mediating PTL endocytosis in RAW264.7 macrophages.

Increasing evidence indicates that mitochondrial damage is closely associated with inflammatory activation and oxidative stress-mediated cytotoxicity (John and Raza, 2022). Mitochondria not only serve as the primary source of intracellular ATP production but also function as critical regulators of ROS generation, calcium homeostasis, and inflammatory signaling (Liu et al., 2025; Oh, 2023). Excessive mitochondrial ROS accumulation and mitochondrial damage are recognized as central pathological events in inflammatory injury (Huang et al., 2023; Thongboontho et al., 2023). Nevertheless, whether PTL induces mitochondrial damage following cellular endocytosis and the molecular mechanisms involved remain unknown.

Therefore, the present study aimed to elucidate the MRC2-mediated endocytic pathway responsible for PTL uptake and further investigate the mechanisms underlying PTL-induced inflammatory toxicity. In particular, we focused on the relationship between PTL endocytosis, mitochondrial damage, and inflammatory responses in RAW264.7 macrophages. Our findings provide new insight into the molecular basis of plant lectin-induced immunotoxicity and establish a theoretical foundation for the safe clinical application and toxicity attenuation of *Pinellia ternata*.

2. Materials and methods

2.1. Plants

Fresh tubers of *Pinellia ternata* (Thunb.) Breit were collected from Yuzhou City, Henan Province, China. The plant material was taxonomically authenticated by Professor Xunhong Liu (Nanjing University of Chinese Medicine, Nanjing, China) as the tubers of *Pinellia ternata* (Thunb.) Breit (Araceae).

2.2. Cell culture

RAW 264.7 murine macrophages, L929 murine fibroblasts, BT549 human breast cancer cells, MCF10A normal human mammary epithelial cells, and HF-91 human fibroblasts were purchased from Procell (CL-0190; CL-0137; CL-0041; CL-0525; CL-0352; Wuhan, China). Cells were cultured in Dulbecco's Modified Eagle Medium (DMEM; PM150210; Procell, Wuhan, China) supplemented with 10% fetal bovine serum (FBS; 164210; Procell, Wuhan, China) and 1% penicillin-streptomycin (C1025; NCM, Suzhou, China) at 37 °C in a humidified incubator containing 5% CO_2. Cells in the logarithmic growth phase were used for all experiments.

2.3. Construction of ATP5B-knockdown stable RAW 264.7 cell lines

Lentivirus-mediated RNA interference was used to establish stable ATP5B-knockdown RAW 264.7 cell lines. Lentiviral particles encoding ATP5B-targeting shRNA (shATP5B-1, −2, −3) or negative control shRNA (shNC) were obtained from Nanjing Cohesion Biosciences Co., Ltd. (Nanjing, China). RAW 264.7 cells were seeded into 6-well plates at a density of 2×10^5 cells/well and cultured overnight. When cell confluence reached approximately 40–50%, cells were infected with lentiviral particles in the presence of polybrene (5 μg/mL). Lentiviral stock solution was added at a multiplicity of infection of 100. After 24 h of infection, the medium was replaced with fresh complete medium. Forty-eight hours later, puromycin (6 μg/mL; ST551; Beyotime, Shanghai, China) was added for selection. The selection medium was refreshed every 2–3 days until stable cell populations were established. Knockdown efficiency was subsequently verified as described in Section 2.13 (Lam et al., 2014; Opaluch et al., 2014; Xian et al., 2021).

2.4. Extraction and purification of PTL

PTL was extracted, separated, and purified following previously established protocols (Wang et al., 2017; Yu et al., 2015). Our previous studies have shown that, the purified protein was identified as PTL by SDS-PAGE, size-exclusion high-performance liquid chromatography (SEC-HPLC), and LC-MS/MS analysis, the purified protein migrated as a single band at approximately 12 kDa on SDS-PAGE, with a purity exceeding 98% as determined by SEC-HPLC; LC-MS/MS analysis further confirmed that the purified protein corresponded to a single-component lectin (Li et al., 2023). The purified PTL was stored at −80 °C until further use.

2.5. Preparation of FITC-labeled PTL

PTL was dissolved in 0.1 M sodium carbonate buffer at a concentration of 2 mg/mL (10,019,261; Sinopharm, Nanjing, China). Fluorescein isothiocyanate (FITC; F7250; Sigma, St. Louis, MO, USA) was dissolved in dimethyl sulfoxide at 1 mg/mL. FITC solution (50 μL) was

added dropwise to each milliliter of PTL solution under continuous stirring. The reaction mixture was incubated in the dark at 4 °C for 8 h. Subsequently, ammonium chloride was added to a final concentration of 50 mM to terminate the reaction, followed by incubation at 4 °C for an additional 2 h. The reaction mixture was dialyzed extensively in the dark to remove unbound FITC, yielding FITC-conjugated PTL (FITC-PTL), which was stored at 4 °C until use.

2.6. Cellular uptake, endocytosis and receptor binding assays

Cytotoxicity of PTL was evaluated by CCK-8 assay. RAW264.7 cells were seeded into 96-well plates and treated with PTL at concentrations of 0, 6.25, 12.5, 25, 50, 100, 200, 400, and 800 μg/mL for 24 h. CCK-8 solution was then added and incubated for 1 h. The OD was measured at 450 nm, and cell viability was calculated as the percentage of the control. Based on these preliminary results (Supplementary Fig. S1) and our previous findings (Yu et al., 2015), 50 μg/mL PTL was selected as the optimal concentration for subsequent experiments.

Cellular uptake: For dynamic uptake monitoring, RAW 264.7 cells were seeded into confocal dishes and imaged by LSCM in xyt mode immediately after the addition of FITC-PTL (50 μg/mL); images were acquired at 2-min intervals over 1 h (Li et al., 2023). For static uptake observation, RAW 264.7 cells, L929 fibroblasts, BT549 cells, MCF10A cells and HF-91 fibroblasts were seeded into confocal dishes and incubated with FITC-PTL (50 μg/mL) at 37 °C for 1 h. Nuclei were stained with Hoechst 33342 (C1025; Beyotime, Shanghai, China) for 10 min, and fluorescence was visualized by LSCM.

Co-localization with clathrin heavy chain (CLTC): RAW 264.7 cells were incubated with FITC-PTL (50 μg/mL) for 1 h, fixed, permeabilized, and incubated overnight at 4 °C with anti-CLTC primary antibody (66487–1; Proteintech, Wuhan, China), followed by fluorophore-conjugated secondary antibody (33208ES60; Yeasen, Shanghai, China) for 2 h at room temperature. Co-localization was visualized by LSCM (Crowther and Schlesinger, 2006).

Pharmacological inhibition of endocytic pathways: RAW 264.7 cells were pretreated with chlorpromazine (50 mM; T22292; TargetMol, Shanghai, China) or indomethacin (100 μM; I106885; Aladdin, Shanghai, China) for 1 h, followed by co-incubation with FITC-PTL (50 μg/mL) for an additional 1 h. Cellular uptake was examined by LSCM (Liu et al., 2024).

Receptor competition assay: RAW 264.7 cells were pretreated with mannan (1 mM; M8730; Solarbio, Beijing, China) or unlabeled PTL for 1 h, followed by incubation with FITC-PTL (50 μg/mL) for 1 h. Fluorescence intensity was analyzed by LSCM and flow cytometry. Molecular docking between MRC2 and PTL was separately performed as described in Section 2.7 (Shibaguchi et al., 2019).

2.7. Homology modeling and molecular docking

Homology modeling of PTL: The sequence of PTL was obtained from GenBank (accession No. ABX47148.1; Supplementary Table 1). A three-dimensional structural model of PTL was generated using the AlphaFold server.

Molecular docking: The three-dimensional structures of mannose receptor (MRC2; UniProt ID: Q9UBG0), ATP5B (UniProt ID: P56480), and PTL (UniProt ID: M1G4Y2) were retrieved from the UniProt database. Molecular docking between PTL and MRC2 or ATP5B was performed using the GRAMM-X server, and the docking results were analyzed using BIOVIA Discovery Studio software (Dassault Systèmes, France).

2.8. Evaluation of inflammatory responses and intracellular ROS

RNA extraction and RT-qPCR: Total RNA was extracted using TRIzon reagent (CW0580SCW3360H; CWBio, Nanjing, China). RNA purity and concentration were determined using a NanoDrop spectrophotometer. Reverse transcription was performed using the HiFiScript All-in-one qRT Master Mix kit (CW3360H; CWBio, Nanjing, China) with 1 μg RNA template. RT-qPCR was performed using specific primers for TNF-α, IL-1β and β-actin (Supplementary Table 2). Relative gene expression was calculated using the $2^{-}\Delta\Delta Ct$ method.

ELISA: Protein concentrations of TNF-α and IL-1β in cell culture supernatants were measured using ELISA kits (ml002095; ml037875; mlbio, Shanghai, China) according to the manufacturer's instructions.

Intracellular ROS: Cells were incubated with DCFH-DA fluorescent probe (S0033S; Beyotime, Shanghai, China) at 37 °C for 30 min. Intracellular ROS levels were analyzed by LSCM and flow cytometry.

2.9. Subcellular co-localization of PTL and assessment of mitochondrial function

PTL-mitochondria co-localization and Mitochondrial function were assessed in RAW 264.7 cells after PTL treatment (50 μg/mL for 1 h).

Co-localization of PTL with mitochondria: After PTL treatment, cells were stained with MitoTracker Red CM-H_2XRos (40740ES50; Yeasen, Shanghai, China) for 20 min, followed by Hoechst 33342 staining for 10 min. Intracellular localization of PTL and mitochondria was observed using LSCM.

Mitochondrial function: After PTL treatment, cells were washed twice with PBS and then incubated with JC-1 probe (5 μg/mL; C2006; Beyotime, Shanghai, China) for mitochondrial membrane potential (ΔΨm) detection, Rhod-2 AM probe (5 μM; 40778ES50; Yeasen, Shanghai, China) for mitochondrial calcium level detection, or MitoSOX Red probe (5 μM; 40776ES50; Yeasen, Shanghai, China) for mitochondrial ROS generation detection at 37 °C for 30 min in the dark. Following probe loading, cells were gently washed three times with PBS to remove extracellular probes. Nuclei were counterstained with Hoechst 33342 for 10 min at room temperature. Fluorescence images were acquired using a laser scanning confocal microscope (LSCM) under identical settings for all groups. ΔΨm was evaluated by the red/green fluorescence ratio of JC-1, while mitochondrial calcium and ROS levels were quantified by mean fluorescence intensity (Minta et al., 1989; Reers et al., 1995; Mukhopadhyay et al., 2007).

2.10. Effects of Mannan and chlorpromazine on PTL-induced inflammatory and mitochondrial responses

RAW 264.7 cells were divided into four groups: Control, PTL alone, Mannan + PTL, and Chlorpromazine + PTL. Cells were pretreated with mannan (1 mM) or chlorpromazine (50 μM) for 1 h, followed by PTL (50 μg/mL) for 1 h. Subsequent assays for inflammatory cytokines, ROS, and mitochondrial function were performed as described in Sections 2.8 and 2.9.

2.11. Identification of PTL-interacting mitochondrial proteins by Co-IP-MS

Co-immunoprecipitation combined with liquid chromatography-tandem mass spectrometry (LC-MS/MS) was employed to identify mitochondrial proteins interacting with PTL. Anti-PTL antibody was conjugated to Protein A/G magnetic beads (HY-K0202; MCE, USA) at 4 °C for 2 h. Mitochondrial protein fractions were isolated from RAW 264.7 cells by differential centrifugation. The antibody-conjugated beads were incubated with mitochondrial lysates for 4 h to capture PTL-interacting proteins. After extensive washing, bound proteins were eluted using glycine buffer, reduced, alkylated, trypsin-digested, desalted, and analyzed by LC-MS/MS. Normal IgG controls were included to eliminate non-specific binding. Raw MS data were processed using MaxQuant software for protein identification.

2.12. Validation of PTL-ATP5B interaction

Co-localization of PTL and ATP5B: RAW 264.7 cells were incubated with FITC-PTL (50 μg/mL) for 1 h, fixed, permeabilized, and incubated overnight at 4 °C with anti-ATP5B primary antibody (17247–1-AP; Proteintech, Wuhan, China), followed by fluorophore-conjugated secondary antibody for 1 h. Images were acquired by LSCM.

MicroScale Thermophoresis (MST): ATP5B was fluorescently labeled using the RED-NHS Protein Labeling Kit (NanoTemper, MO-L011). Labeled ATP5B was diluted to 100 nM in HEPES buffer (50 mM, pH 7.4, 0.05% Tween-20). PTL was serially diluted 2-fold from 225 nM to 6.9 pM and mixed with equal volumes of labeled ATP5B. Binding signals were measured using a NanoTemper Monolith X instrument. Data were fitted to a sigmoidal binding model using MO.Affinity Analysis software to calculate the Kd value (mean ± SD from three technical replicates; performed by Nanjing Ruiyuan Biotechnology Co., Ltd.).

2.13. Verification of ATP5B-knockdown stable RAW 264.7 cell lines

Knockdown efficiency of the stable cell lines constructed in Section 2.3 was verified by Western blot and RT-qPCR. For Western blot, RAW 264.7 cells were divided into four groups: shNC, shATP5B-1, shATP5B-2, and shATP5B-3. Total cellular proteins were extracted using RIPA lysis buffer (WB3100; NCM, Suzhou, China) supplemented with protease and phosphatase inhibitors. Protein samples were separated by SDS-PAGE and transferred onto PVDF membranes. Membranes were blocked and incubated overnight at 4 °C with primary antibodies against ATP5B (1:3000) and GAPDH (1:50000; 60,004–1-Ig; Proteintech, Wuhan, China), followed by HRP-conjugated secondary antibodies (A0208; Beyotime, Shanghai, China). Protein bands were visualized using enhanced chemiluminescence and quantified by densitometric analysis. For RT-qPCR analysis, total RNA extraction and quantitative PCR procedures were performed as described in Section 2.8, using primers specific for ATP5B and GAPDH (Supplementary Table 3).

2.14. Effects of ATP5B knockdown on PTL-induced inflammatory and mitochondrial responses

Stable shNC and shATP5B cells (using the shRNA construct with the highest knockdown efficiency selected from Section 2.13) were divided into three groups: shNC, shNC + PTL, and shATP5B + PTL. Cells were treated with PTL (50 μg/mL) for 1 h. Subsequent assays for inflammatory cytokines, ROS, and mitochondrial function were performed as described in Sections 2.8 and 2.9.

2.15. Statistical analysis

All statistical analyses were performed using GraphPad Prism 8.0 software (GraphPad Software, Boston, MA, USA). Data are presented as mean ± standard deviation (SD). Statistical significance among multiple groups was analyzed using one-way analysis of variance (ANOVA), followed by appropriate post hoc tests. All experiments were independently repeated at least three times, and technical replicates were performed in triplicate. A value of $P < 0.05$ was considered statistically significant.

3. Results

3.1. PTL cellular entry is primarily mediated by Clathrin-dependent endocytosis

Time-lapse laser scanning confocal microscopy revealed that PTL progressively accumulated at the plasma membrane of RAW264.7 cells and subsequently underwent membrane invagination and cytoplasmic endocytosis in a time-dependent manner. Endocytosed PTL appeared as dynamic punctate vesicular structures exhibiting continuous intracellular movement, indicating active endocytic trafficking (Fig. 1A, Supplementary video S1).To exclude potential interference arising from the intrinsic phagocytic activity of macrophages, PTL uptake was further examined in non-phagocytic cell lines, including L929 murine fibroblasts, BT549 human breast cancer cells, MCF10A human mammary epithelial cells, and HF-91 human fibroblasts. Confocal imaging demonstrated that PTL was efficiently endocytosed in all tested cell types, indicating that PTL uptake was not restricted to professional phagocytes and suggesting that PTL possessed intrinsic endocytic capability across multiple cell lineages (Fig. 1B). These findings collectively indicated that PTL initially bound to the cell membrane and subsequently entered cells through an active endocytic process.

To further characterize the endocytic pathway involved in PTL endocytosis, cells were pretreated with chlorpromazine, an inhibitor of clathrin-mediated endocytosis, or indomethacin, an inhibitor of caveolae-dependent endocytosis. Chlorpromazine markedly reduced intracellular fluorescence intensity of FITC-PTL, whereas indomethacin exerted minimal effects, indicating that PTL endocytosis primarily depended on clathrin-mediated endocytosis (Fig. 1C).To further validate this mechanism, co-localization analysis between PTL and clathrin heavy chain (CLTC) was performed. Confocal microscopy demonstrated substantial intracellular co-localization of PTL with CLTC, providing additional evidence that PTL entry into RAW264.7 cells was predominantly mediated through the clathrin-dependent endocytic pathway (Fig. 1D).

3.2. Structural characterization and model validation of PTL based on AlphaFold

Five structural models of PTL were generated using AlphaFold, and the model with the highest confidence score was selected for subsequent analyses (Fig. 2A). The predicted three-dimensional structure of PTL is available in the UniProt database under the accession number M1G4Y2. The stereochemical quality of the predicted PTL structure was further evaluated using Ramachandran plot analysis. Most amino acid residues were distributed within the favored regions (dark-colored areas) or additionally allowed regions (light-colored areas), whereas only phenylalanine 144 (Phe144) and alanine 255 (Ala255) were located outside the allowed regions, indicating acceptable stereochemical quality of the predicted structure (Fig. 2B).

Per-residue confidence scores (pLDDT) generated by AlphaFold demonstrated that the overall pLDDT score of PTL was 82.38, indicating high confidence in the predicted structure. Structural reliability was further assessed using predicted aligned error (PAE) analysis. The PAE matrix exhibited distinct block-like regions along the diagonal, with intra-domain PAE values predominantly below 5 Å, indicating high reliability of individual domain folding. In contrast, inter-domain PAE values exceeding 15 Å suggested potential flexibility between domains. Lower PAE values reflected higher reliability of the relative positions between structural domains (Fig. 2C). Collectively, these analyses demonstrated that the AlphaFold-predicted PTL structure possessed favorable stereochemical quality and structural reliability.

3.3. The binding of PTL to cell membranes is primarily mediated by the mannose receptor C-type 2

To investigate whether the binding of PTL to the cell membrane was mediated by the mannose receptor C-type 2 (MRC2), molecular docking analysis between PTL and MRC2 was performed using the GRAMM-X platform. The highest-scoring docking conformation was subsequently analyzed using PDBePISA (Fig. 3A). PTL and MRC2 exhibited an interaction interface area of 2682.7 $Å^2$ and a binding free energy of −25.2 kcal/mol (Table 1), indicating strong binding affinity between the two proteins.

To validate the molecular docking results, flow cytometry and laser scanning confocal microscopy were employed to assess changes in

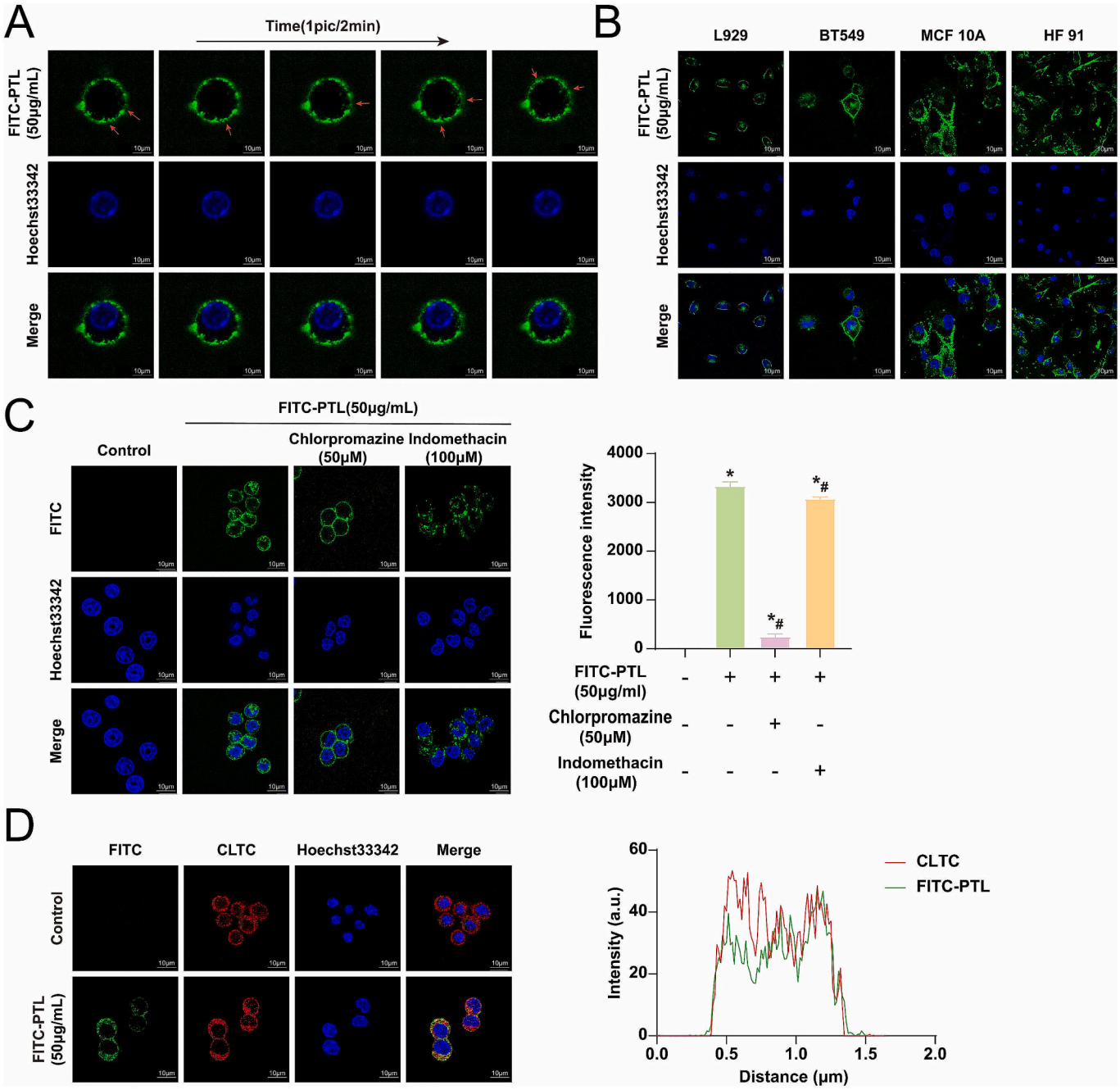


Fig. 1. PTL employed clathrin-mediated endocytosis for cellular entry. (A) Dynamic endocytosis of PTL was visualized in RAW 264.7 macrophages. (B) Endocytosis of PTL was examined in different cell lines, including L929, BT549, MCF-10 A, and HF91. (C) The effects of endocytosis inhibitors (chlorpromazine and indomethacin) on PTL endocytosis were assessed. Quantitative analysis of the effects of chlorpromazine and indomethacin on the endocytic activity of PTL in immunofluorescence staining was shown in the right panel (Statistical significance was defined as $p < 0.05$, $^{*}p < 0.05$ vs. blank control group; $^{\#}p < 0.05$ vs. FITC-PTL group). (D) Co-localization of PTL with CLTC in RAW 264.7 macrophages. Quantitative analysis of co-localization between PTL and CLTC in immunofluorescence staining was shown in the right panel. Nuclei were counterstained with Hoechst 33342 (blue). Scale bar, 10 μm. Values were presented as mean ± standard deviation. All experiments included $n = 3$ independent biological replicates per group, with technical triplicates in each. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

cellular fluorescence intensity. Flow cytometry revealed that the binding of FITC-PTL to RAW 264.7 cells resulted in a 65.5-fold increase in fluorescence intensity compared to the control group (Fig. 3B). Pretreatment with mannan, a competitive inhibitor of MRC2, reduced the fluorescence intensity by 25.7%, indicating that MRC2 contributed to PTL binding. Following pretreatment with unlabeled PTL, the fluorescence intensity was reduced by 53.3%, representing an additional 27.6% inhibition beyond that achieved by mannan, suggesting the involvement of additional membrane receptors in PTL binding.

Consistent with the flow cytometry results, laser scanning confocal microscopy revealed that FITC-PTL fluorescence was predominantly localized on the cell surface and within the cytoplasm (Fig. 3C). Quantitative analysis showed that pretreatment with mannan reduced fluorescence intensity by 49.8%, while pretreatment with unlabeled PTL reduced it by 76.5%, with an additional 26.7% inhibition beyond mannan treatment. These results further confirmed that, in addition to binding to MRC2, PTL could also interact with other receptors on the cell membrane.

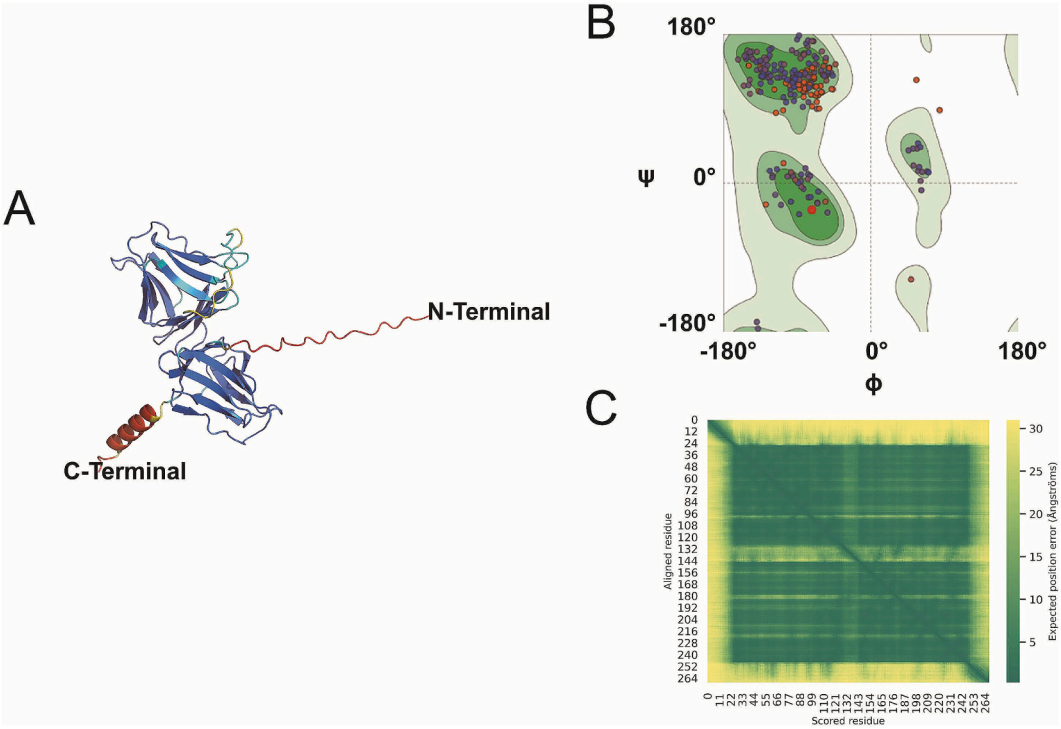


Fig. 2. AlphaFold prediction of the three-dimensional structure of PTL. (A) 3D structure prediction of PTL. **(B)** Ramachandran plot of the predicted structure of PTL. (C) Confidence score of the predicted structure of PTL.

Taken together, these results indicated that PTL binding to the cell membrane was mediated by multiple receptors, among which MRC2 was a significant binding site.

3.4. PTL induces inflammation in RAW 264.7 cells via mannose receptor C-type 2-mediated endocytosis

Compared with the control group, PTL treatment significantly increased the mRNA expression levels of TNF-α and IL-1β in RAW264.7 cells. Pretreatment with either mannan or chlorpromazine markedly attenuated PTL-induced upregulation of TNF-α and IL-1β, indicating that inhibition of MRC2 binding or clathrin-mediated endocytosis effectively suppressed PTL-induced inflammatory gene expression (Fig. 4A). To further validate these findings, the protein levels of TNF-α and IL-1β in culture supernatants were measured by ELISA. Consistent with the RT-qPCR results, both mannan and chlorpromazine significantly reduced PTL-induced cytokine secretion (Fig. 4B), confirming that PTL-triggered inflammatory responses were dependent on MRC2-mediated endocytosis. Intracellular ROS production was subsequently evaluated using confocal microscopy and flow cytometry. PTL treatment markedly enhanced intracellular fluorescence intensity, indicating substantial ROS accumulation. In contrast, pretreatment with mannan or chlorpromazine significantly reduced ROS generation (Fig. 4C). Flow cytometric analysis further demonstrated a leftward shift in fluorescence intensity following inhibition of MRC2 binding or endocytosis, confirming attenuation of PTL-induced oxidative stress (Fig. 4D). These results demonstrated that PTL induced ROS production and inflammatory activation in RAW 264.7 cells primarily through MRC2-mediated, clathrin-dependent endocytosis.

3.5. PTL-induced mitochondrial damage in RAW264.7 cells is dependent on mannose receptor C-type 2-mediated endocytosis

Confocal microscopy revealed substantial co-localization of PTL with mitochondria following cellular endocytosis. In contrast, pretreatment with mannan or chlorpromazine markedly reduced mitochondrial co-localization of PTL, indicating that PTL was transported to mitochondria following MRC2-mediated endocytosis (Fig. 5A). Mitochondrial membrane potential was assessed using JC-1 staining. Compared with the control group, PTL treatment significantly decreased the red/green fluorescence ratio, indicating reduction of mitochondrial membrane potential. However, pretreatment with mannan or chlorpromazine significantly alleviated PTL-induced mitochondrial membrane potential reduction (Fig. 5B). Mitochondrial calcium levels were subsequently evaluated using Rhod-2 AM staining. PTL treatment markedly increased mitochondrial calcium accumulation compared with the control group, whereas pretreatment with mannan or chlorpromazine significantly attenuated this increase (Fig. 5C). Similarly, mitochondrial ROS production detected by MitoSOX Red staining was significantly elevated following PTL treatment. Inhibition of MRC2 binding or clathrin-mediated endocytosis substantially reduced mitochondrial ROS accumulation induced by PTL (Fig. 5D). These results demonstrated that PTL induced mitochondrial damage in RAW 264.7 cells through MRC2-mediated endocytosis.

3.6. Identification of ATP5B as the key mitochondrial target underlying PTL toxicity

To identify mitochondrial proteins interacting with PTL, Co-IP-MS/MS analysis was performed. These results revealed that 268 proteins were pulled down in the PTL group, whereas only 18 proteins were pulled down in the group treated with heat-denatured PTL (Fig. 6A). Further analysis of the proteins pulled down in the PTL group identified 33 mitochondrial-related proteins (Fig. 6B). Several mitochondrial-interacting proteinswere identified, including Atp5b (ATP synthase subunit beta), Hspd1 (60 kDa heat shock protein), Slc25a4 (ADP/ATP translocase), Oxct1 (succinyl-CoA: 3-ketoacid coenzyme A transferase 1), Ogdh (2-oxoglutarate dehydrogenase), etc. (Table 2). Among the ten mitochondrial proteins with the highest enrichment intensities, ATP5B exhibited the strongest enrichment.

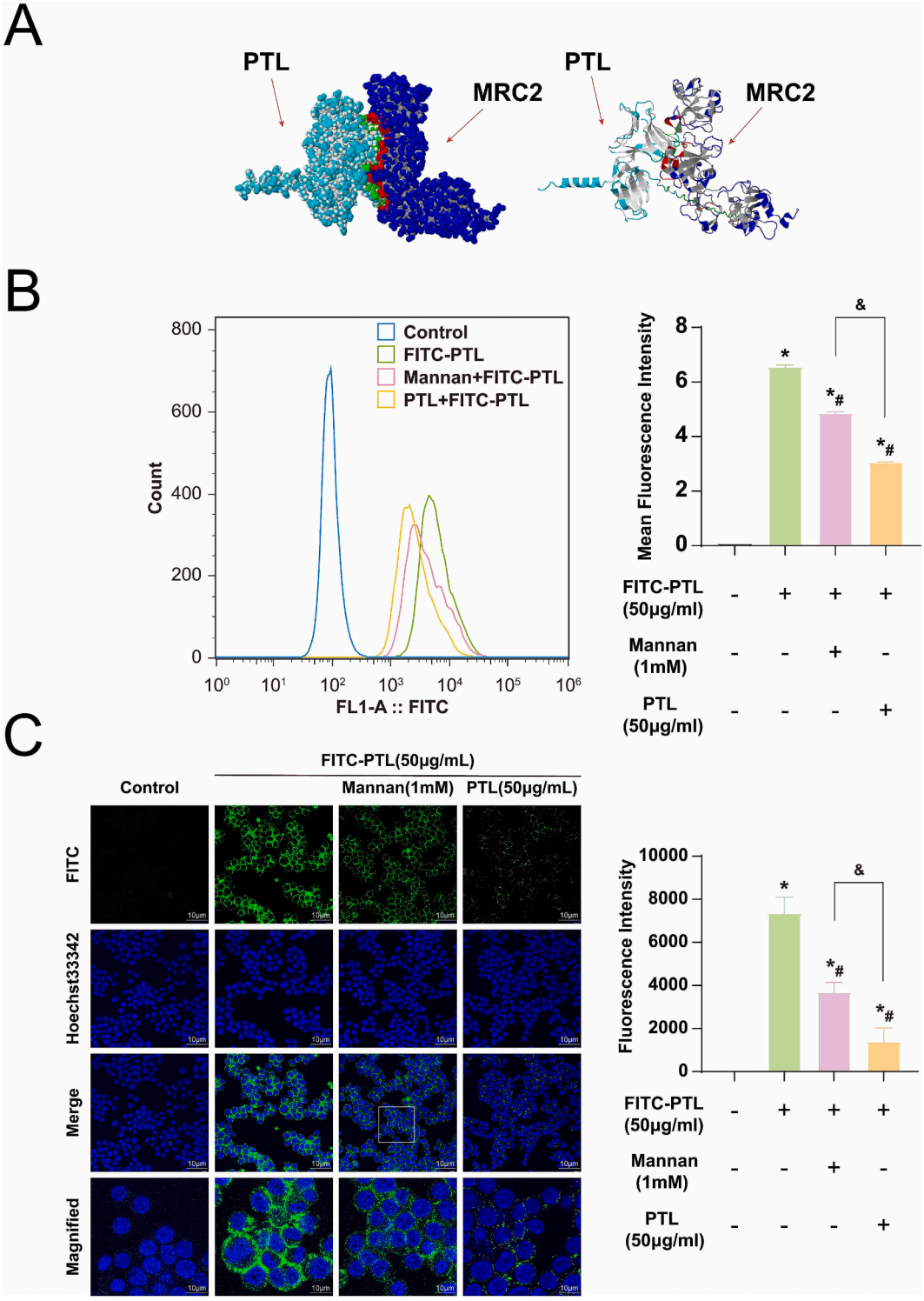


Fig. 3. Binding of PTL to the cell membrane was mediated by the mannose receptor. (A) Molecular docking analysis between PTL and MRC2 was performed to predict their interaction. The 3D structures of the proteins were obtained from the UniProt and RCSB PDB databases, and the docking simulation was performed using BIOVIA Discovery Studio. (B) Binding of FITC-PTL (green) to cells in different treatment groups was assessed by flow cytometry. The quantitative analysis of FITC-PTL fluorescence intensity was shown in the right panel. (C) The binding of FITC-PTL (green) to cells across different treatment groups was observed using laser scanning confocal microscopy. Nuclei were counterstained with Hoechst 33342 (blue). Scale bar, 10 μm. The right panel showed the quantitative analysis of PTL endocytosis into cells. Values were presented as mean ± standard deviation. Statistical significance was defined as $p < 0.05$, $^{*}p < 0.05$ vs. control; $^{\#}p < 0.05$ vs. FITC-PTL group; and $^{\&}p < 0.05$ vs. Mannan + FITC-PTL group. All experiments included n = 3 independent biological replicates per group, with technical triplicates in each. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Molecular docking analysis of PTL with MRC2.

Protein	Atoms	Interfacing Residues	SASA (Å^2)	Hydrogen Bonds	BSA (Å^2)	ΔG (kcal/mol)
PTL	258	81	21,496	20	2682.7	−25.2
MRC2	284	53	16,380			

(SASA: Solvent Accessible Surface Area; BSA: Buried Surface Area; ΔG: Binding Free Energy)

These findings suggested that PTL might interact with ATP5B. Molecular docking was performed to validate this hypothesis. As shown in (Fig. 6C and Table 3), the interaction area between PTL and ATP5B was 1957.7 Å^2, with a free binding energy of −3.1 kcal/mol. These results supported the feasibility of PTL-ATP5B binding. The interaction between PTL and ATP5B was further confirmed using laser scanning confocal microscopy to observe their co-localization. PTL exhibited clear co-localization with ATP5B, demonstrating that PTL could interact with ATP5B (Fig. 6D).

To further quantitatively evaluate the direct binding affinity between PTL and ATP5B, microscale thermophoresis (MST) binding assays were performed. Fluorescently labeled ATP5B protein was incubated with serially diluted PTL, and the binding signals were measured. As shown in (Fig. 6E), the normalized fluorescence signals exhibited a typical sigmoidal dose-dependent change with increasing PTL concentrations. Fitting the data to a sigmoidal binding model yielded a dissociation constant (Kd) of 3.52 ± 0.31 μM for the PTL-ATP5B interaction. This result quantitatively confirmed a direct interaction between PTL and ATP5B, further supporting that ATP5B was a potential direct target of PTL in mitochondria.

3.7. Knockdown of ATP5B rescues PTL-induced inflammatory responses

To investigate whether ATP5B mediated PTL-induced inflammatory responses, lentivirus-mediated transfection was employed to knock down ATP5B in macrophages. Among the three shRNAs targeting ATP5B, shATP5B-2 exhibited the highest knockdown efficiency and was thus selected for subsequent experiments (Fig. 7A-C). Intracellular ROS production was evaluated by laser scanning confocal microscopy and flow cytometry. Compared with the shNC group, PTL treatment significantly enhanced ROS release; however, ATP5B knockdown markedly suppressed PTL-induced ROS accumulation (Fig. 7D and E). The expression levels of TNF-α and IL-1β were subsequently examined by RT-qPCR and ELISA. In the shNC group, PTL treatment significantly upregulated both the mRNA expression and protein secretion of TNF-α and IL-1β. In contrast, ATP5B knockdown dramatically inhibited PTL-induced production of these inflammatory cytokines (Fig. 7F and G). These results indicated that ATP5B knockdown exerted a protective effect by reducing the available binding sites for PTL, thereby blocking PTL-triggered oxidative stress and inflammatory signaling. Collectively, these findings confirmed that PTL exerted its pro-inflammatory effects in RAW264.7 macrophages through interaction with ATP5B.

3.8. Knockdown ATP5B alleviates PTL-induced mitochondrial damage

To further determine whether ATP5B mediated PTL-induced mitochondrial injury, we assessed mitochondrial membrane potential, mitochondrial calcium levels, and mitochondrial ROS production following ATP5B knockdown. Compared with the shNC group, PTL treatment significantly reduced mitochondrial membrane potential, as evidenced by a decreased JC-1 red/green fluorescence ratio. Notably, ATP5B knockdown markedly alleviated the PTL-induced reduction in mitochondrial membrane potential (Fig. 8A). Furthermore, relative to the shNC group, PTL treatment significantly increased mitochondrial calcium accumulation. However, ATP5B knockdown substantially attenuated PTL-induced mitochondrial calcium overload (Fig. 8B). Similarly, PTL treatment significantly elevated mitochondrial ROS production compared with the shNC group, whereas ATP5B knockdown markedly suppressed PTL-induced mitochondrial ROS generation (Fig. 8C). Collectively, these findings demonstrated that ATP5B knockdown alleviated PTL-induced mitochondrial injury via diminishing the binding sites available for PTL. These results further corroborated that PTL induced mitochondrial damage in RAW264.7 macrophages through interaction with ATP5B.

4. Discussion

Plant lectins are a class of non-immune proteins characterized by specific carbohydrate-binding capacities and diverse biological activities, which make them suitable for broad applications in both medical and agricultural fields (Nascimento et al., 2020). Since their initial discovery in castor beans (Ting et al., 2009), numerous lectins have been identified, isolated, and purified from a wide range of organisms, including animals, plants, viruses, and bacteria (Biscotti et al., 2025; Zajaczkowski et al., 2025). In addition to the potent cytotoxicity and ribosome-inactivating activities exhibited by certain lectins, some lectins also possess pronounced pro-inflammatory toxicity (Yu et al., 2015).

Pinellia ternata is a commonly used traditional Chinese medicinal herb with acrid and warm properties, primarily acting through the spleen, stomach, and lung meridians. Its therapeutic effects include drying dampness and resolving phlegm, and it has been clinically applied to treat productive cough, dyspnea, and dizziness associated with phlegm retention (Liang et al., 2025). However, its tubers contain irritating components that cause mucosal inflammation (Zou et al., 2023), although this toxicity can be attenuated through processing. Previous studies demonstrated that *Pinellia ternata* lectin (PTL), the principal toxic component of *Pinellia ternata*, exhibited strong membrane-binding activity (Shi et al., 1994; Shih et al., 1992) and was progressively endocytosed into cells over time, suggesting the involvement of an active endocytic process. Several plant lectins, including ricin and *Concanavalin* A (ConA), have been reported to enter mammalian cells via receptor-mediated endocytosis, thereby inducing cytotoxicity and inflammatory signaling activation (Gao et al., 2020; Ginini et al., 2022). Previous investigations further revealed that PTL utilized calcium oxalate raphides to penetrate tissues and aggravate local inflammatory responses (Zhu et al., 2012). Notably, the inflammatory process induced by these toxic raphides is mainly mediated by macrophages, as demonstrated in mouse peritoneal macrophage culture models (Zhao et al., 2013). These findings suggest that the pro-inflammatory effect of PTL requires physical penetration by raphides to access tissues, where macrophages are subsequently activated to initiate inflammatory responses. Interestingly, this inflammatory effect is not restricted to the digestive system, as intraperitoneal and ocular administration also elicited robust inflammation in vivo (Zhu et al., 2012), indicating that PTL possesses tissue-independent pro-inflammatory potential. However, whether PTL can reach its target tissues intact after oral ingestion remains unclear. Although heating or ginger extract has been shown to reduce PTL content in raphides (Liu et al., 2023), systematic studies on the proteolytic stability of PTL in the digestive tract are currently lacking, warranting further investigation. Nevertheless, despite these findings, the mechanism through which PTL subsequently entered cells to trigger intracellular inflammatory signaling remained unclear, thereby limiting both the clinical application of *Pinellia ternata* and the utilization of PTL as a bioactive tool protein in biomedical research.

Endocytosis is a fundamental cellular transport process through which extracellular substances, including proteins, lipids, pathogens, and macromolecular complexes, are internalized into cells. It encompasses several uptake pathways, including clathrin-mediated endocytosis, caveolae-dependent endocytosis, macropinocytosis, and phagocytosis (Lin and Shi, 2019). Among these pathways, clathrin-mediated endocytosis represents one of the most extensively

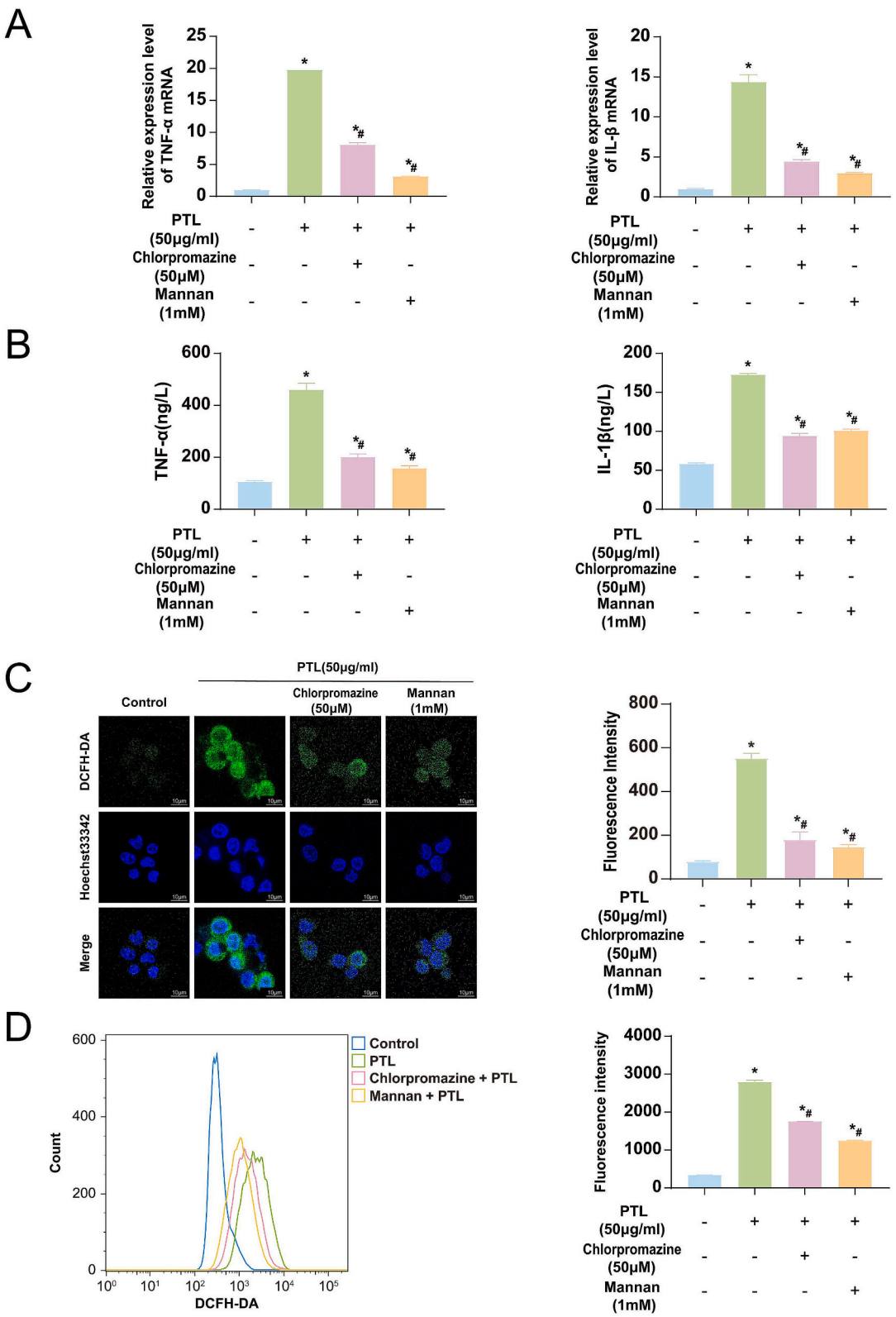


Fig. 4. PTL induced inflammatory responses in RAW 264.7 cells through mannose receptor-mediated endocytosis. (A) Relative mRNA expression levels of the inflammatory cytokines TNF-α and IL-1β in RAW 264.7 cells treated with PTL, quantified using RT-qPCR. (B) Levels of TNF-α and IL-1β inflammatory cytokines in RAW 264.7 cells treated with PTL, as determined using ELISA ($n = 6$ for ELISA). (C) Intracellular ROS levels were detected using laser scanning confocal microscopy. Nuclei were counterstained with Hoechst 33342 (blue). Scale bar, 10 μm. The quantitative analysis of ROS fluorescence intensity was shown in the right panel. (D) Intracellular ROS levels were measured by flow cytometry. The quantitative analysis of ROS fluorescence intensity was shown in the right panel. $^{*}p < 0.05$ vs. control group; $^{\#}p < 0.05$ vs. PTL group. Values were presented as mean ± standard deviation. All experiments included n = 3 independent biological replicates per group, with technical triplicates in each. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

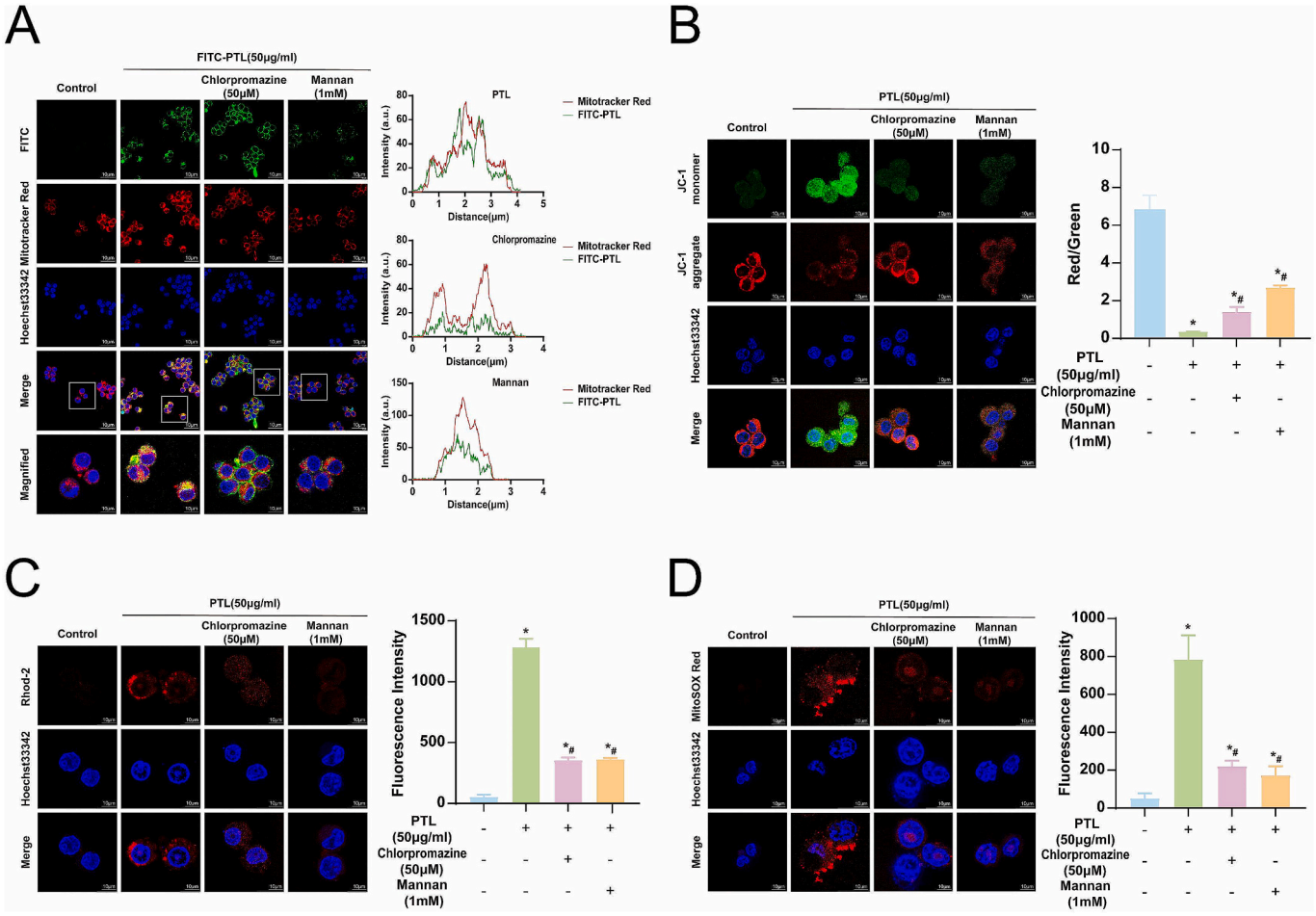
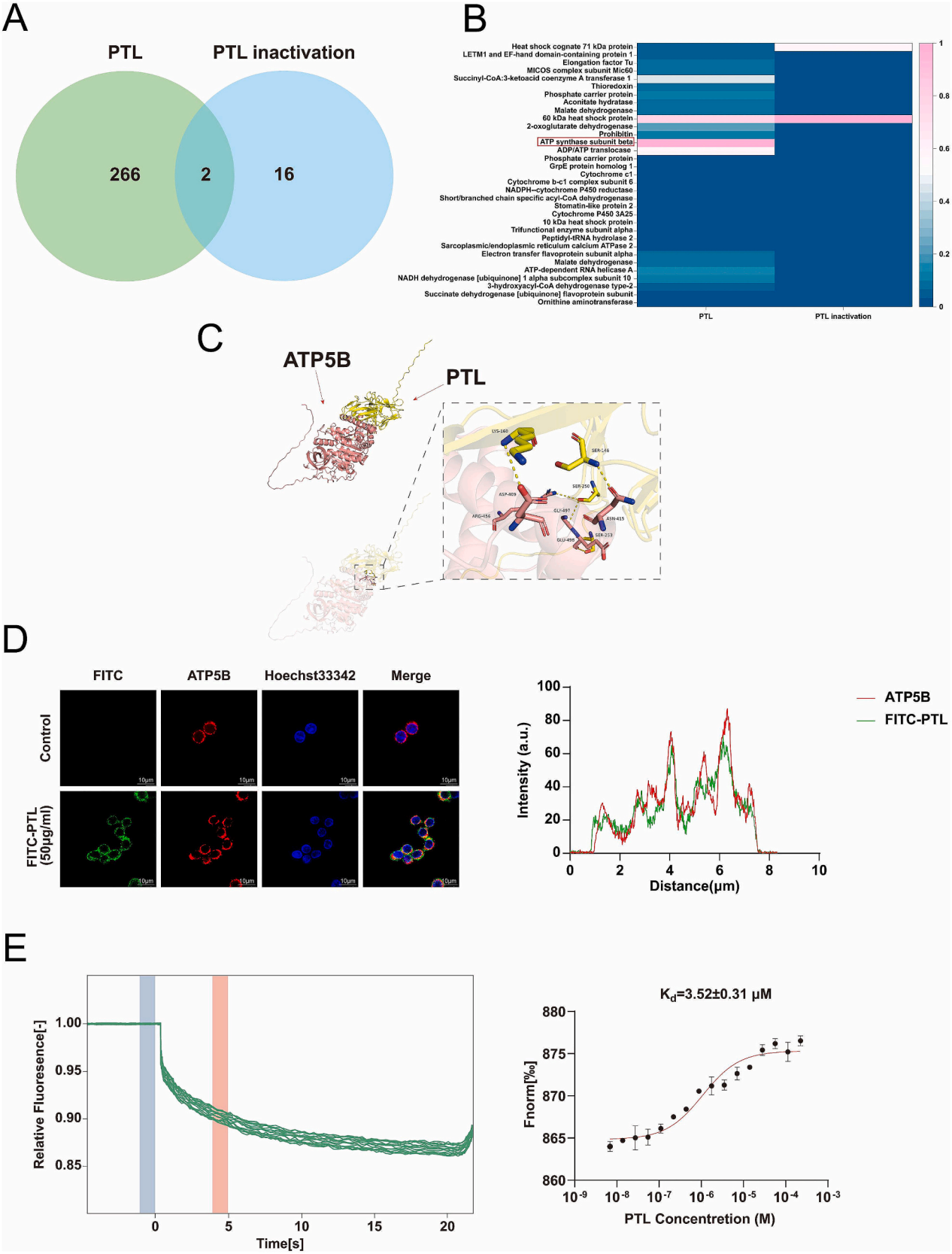


Fig. 5. Mitochondrial damage in RAW 264.7 cells was induced by PTL via mannose receptor-mediated endocytosis. (A) Mitochondria in RAW 264.7 cells were labeled with MitoTracker Red (red), and PTL was labeled with FITC (green). Single-channel and merged images were acquired using laser scanning confocal microscopy. Results of the fluorescence intensity analysis for PTL-mitochondria co-localization were presented in the right panel. (B) After staining different treatment groups with JC-1 dye, images of the JC-1 monomer (green) and JC-1 aggregate (red) channels were acquired using laser scanning confocal microscopy. The quantitative analysis results were shown in the right panel. (C) Detection of mitochondrial calcium levels in different treatment groups with Rhod-2 (red) staining using laser scanning confocal microscopy. The quantitative analysis results were shown in the right panel. (D) Representative images of MitoSOX Red staining (red) showing mitochondrial superoxide levels in different treatment groups, as acquired using laser scanning confocal microscopy. Nuclei were counterstained with Hoechst 33342 (blue). Scale bar, 10 μm. The quantitative analysis results were shown in the right panel. $^{*}p < 0.05$ vs. control group; $^{\#}p < 0.05$ vs. PTL group. Values were presented as mean ± standard deviation. All experiments included n = 3 independent biological replicates per group, with technical triplicates in each. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

characterized receptor-dependent transport mechanisms. During this process, adaptor protein complexes recruit clathrin molecules to the plasma membrane, leading to the formation of clathrin-coated pits and subsequent vesicle budding (Ju et al., 2020). Dynamin-mediated membrane scission subsequently releases intracellular vesicles, which fuse with early endosomes for cargo trafficking (Link et al., 2021). Importantly, some lectins preferentially utilize clathrin-mediated pathways for cellular uptake because lectin-receptor interactions promote receptor clustering on the plasma membrane, thereby facilitating adaptor protein recruitment and coated-pit assembly (Chang et al., 1993; Mettlen et al., 2018). In the present study, time-lapse confocal imaging demonstrated that PTL underwent progressive endocytosis in RAW264.7 macrophages, whereas PTL uptake was also observed in multiple non-phagocytic cell lines, indicating that its uptake was not restricted to professional phagocytes. Chlorpromazine, a classical inhibitor of clathrin-mediated endocytosis that disrupts clathrin lattice assembly and adaptor redistribution (Daniel et al., 2015), markedly inhibited PTL uptake. In contrast, indomethacin, a pharmacological modulator of caveolae/lipid raft-associated internalization pathways (Filippini and Alessio, 2020), showed minimal inhibitory effects. Furthermore, PTL exhibited clear co-localization with CLTC (clathrin heavy chain), a core structural protein required for clathrin-mediated endocytosis, providing further evidence for the involvement of this pathway.

The fate of these vesicles and their downstream biological effects largely depend on the specific ligands recognized by surface receptors. Therefore, identifying receptor specificity in endocytosis is essential for elucidating downstream signaling pathways. Although PTL exhibits strong membrane-binding capacity, the specific receptor mediating its endocytosis has not been determined. Receptor-mediated endocytosis represents a coordinated cellular process in which selected extracellular proteins or peptides are first bound to specific surface receptors and are subsequently endocytosed, playing essential roles in cellular growth, nutrition, and differentiation (Rajaram et al., 2017). Previous studies establish that clathrin-mediated endocytosis facilitates the endocytosis of various receptors, including the mannose receptor (Ginini et al., 2022). Our investigations demonstrated that PTL stimulation upregulated MRC2 expression in RAW 264.7 macrophages (Wang, 2019). As a member of the mannose receptor family, MRC2 (CD280/Endo180) shares structural homology with the classical mannose receptor MRC1 (CD206) (East and Isacke, 2002). Unlike the well-characterized MRC1



(caption on next page)

Fig. 6. Interaction of PTL with ATP5B. (A) Venn diagram of proteins from RAW 264.7 macrophage lysates that were identified to interact with PTL using co-immunoprecipitation-mass spectrometry (Co-IP-MS). (B) Heatmap of the expression of identified mitochondrial proteins that interacted with PTL in RAW 264.7 macrophage lysates. Expression levels are presented in a colour gradient, with pink indicating high expression and blue indicating low expression. Columns correspond to individual proteins, and rows represent the PTL group versus the boiled, inactivated PTL group. (C) Molecular docking was performed to predict the interactions of PTL with ATP5B, using their respective 3D structures obtained from the UniProt and RCSB PDB databases and simulated using BIOVIA Discovery Studio. (D) The co-localization of PTL and ATP5B was observed using laser scanning confocal microscopy. The quantitative results of PTL-ATP5B co-localization were presented in the right panel. (E) Microscale thermophoresis (MST) binding assays were performed to determine the direct interaction between PTL and ATP5B. The MST trace (left panel) depicts the time-dependent fluorescence intensity changes following infrared laser activation. Normalized differential fluorescence was plotted against PTL concentrations to generate the binding curve (right panel). Scale bar, 10 μm. All experiments included n = 3 independent biological replicates per group, with technical triplicates in each. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Mitochondrial-related proteins pulled down by PTL.

Gene	Protein	PTL (Mean intensity)	PTL inactivation (Mean intensity)	Enrichment
Atp5b	ATP synthase subunit beta	16,918,465	0	16,918,465
Hspd1	60 kDa heat shock protein	14,366,110	922,150	13,443,960
Slc25a4	ADP/ATP translocase	9,968,315	0	9,968,315
Oxct1	Succinyl-CoA:3-ketoacid coenzyme A transferase 1	7,532,080	0	7,532,080
Ogdh	2-oxoglutarate dehydrogenase	4,010,520	0	4,010,520
Dhx9	ATP-dependent RNA helicase A	2,814,095	0	2,814,095
Ndufa10	NADH dehydrogenase	2,382,165	0	2,382,165
Slc25a3	Phosphate carrier protein	2,367,000	0	2,367,000
Phb	Prohibitin	2,181,950	0	2,181,950
Mdh2	Malate dehydrogenase	1,734,950	0	1,734,950

Table 3
Molecular docking analysis of PTL with ATP5B.

Protein	Atoms	Interfacing Residues	SASA ($Å^2$)	Hydrogen Bonds	BSA ($Å^2$)	ΔG (kcal/mol)
PTL	191	49	28,040	5	1957.7	−3.1
ATP5B	167	57	16,803			

(SASA: Solvent Accessible Surface Area; BSA: Buried Surface Area; ΔG: Binding Free Energy)

(CD206), which recognizes terminal mannose or fucose residues on glycoproteins and mediates the endocytosis of various plant lectins, including ConA and GNA (Van der Zande et al., 2021; Zlotnikov and Kudryashova, 2022), MRC2 predominantly functions in collagen internalization and extracellular matrix turnover (Melander et al., 2015). Considering that PTL is a mannose-binding lectin, we therefore evaluated whether MRC2, in addition to MRC1, acts as a receptor mediating PTL endocytosis in macrophages. To provide a structural basis for further analysis, we predicted the three-dimensional structure of PTL using AlphaFold. The obtained model (UniProt accession number M1G4Y2) was evaluated as highly reliable, confirming that the predicted structure is reasonable for subsequent molecular docking studies. Molecular docking analysis further demonstrated favorable interactions between PTL and the mannose receptor C-type 2 (MRC2), while competitive inhibition using mannan significantly attenuated PTL binding and endocytosis efficiency. Collectively, these findings established that PTL entered cells via MRC2-mediated, clathrin-dependent endocytosis.

Oxidative stress is widely recognized as a critical mechanism underlying inflammatory tissue injury (Wang et al., 2025). Helmut Sies first introduced the concept of oxidative stress to describe the imbalance between oxidant production and antioxidant defense systems (Toyokuni, 2016), which was subsequently refined as "an imbalance between oxidants and antioxidants in favor of oxidants, leading to disruption of redox signaling and molecular damage" (Sies et al., 2017). Previous studies demonstrated that PTL stimulation induced substantial ROS generation in RAW264.7 macrophages, thereby activating NF-κB signaling pathways and promoting TNF-α and IL-1β secretion (Mao et al., 2017). Given that PTL was progressively endocytosed into RAW264.7 macrophages over time (Li et al., 2023), we next investigated whether PTL induced inflammation through endocytic mechanisms. Supporting this premise, previous reports confirmed that certain lectins exhibited pro-inflammatory toxicity mediated by endocytosis, leading to cell death, necrotic pathology, and significant cytokine release (Battelli et al., 2001; Benjamin et al., 1997). In the present study, PTL markedly enhanced intracellular ROS accumulation and inflammatory cytokine production, whereas pharmacological inhibition of mannose receptor C-type 2 (MRC2) or clathrin-mediated endocytosis significantly attenuated these effects. Collectively, these findings indicated that PTL elicited inflammatory activation through MRC2-mediated endocytosis in RAW 264.7 macrophages.

Under physiological conditions, reactive oxygen species (ROS) serve as essential redox signaling molecules that maintain normal cellular functions through tightly regulated signal transduction pathways (Vallières et al., 2023). At moderate levels, ROS, particularly hydrogen peroxide and superoxide anions, act as important second messengers involved in the regulation of cell proliferation, differentiation, vascular homeostasis, and metabolic processes (Chen et al., 2020; Yang et al., 2018). These physiological functions rely on a finely balanced redox system in which antioxidant defenses tightly control intracellular ROS levels to prevent oxidative damage. However, disruption of redox homeostasis results in excessive ROS accumulation and subsequent cellular injury (Lee, 2024). Elevated ROS levels induce irreversible oxidative modifications of biomolecules, including lipid peroxidation, protein carbonylation, and DNA damage (Endale et al., 2023). These alterations further activate stress-related signaling pathways, such as JNK, p38, and NF-κB, promote mitochondrial dysfunction, and ultimately trigger multiple forms of cell death, including apoptosis, necrosis, and ferroptosis (Golovach et al., 2023; Olaru et al., 2025).

Mitochondria are the principal intracellular source of ROS and play central roles in regulating cellular redox homeostasis and inflammatory signaling. Excessive ROS accumulation promotes mitochondrial dysfunction, which further amplifies ROS production through ROS-induced ROS release mechanisms, thereby forming a self-perpetuating cycle of oxidative injury (Zahan et al., 2020). Mitochondrial integrity is critically dependent on the maintenance of membrane potential (ΔΨm), calcium homeostasis, and ROS equilibrium (Giorgi et al., 2018; Weindel et al., 2022), with ΔΨm dissipation serving as a hallmark of oxidative stress-induced mitochondrial injury (Makinde et al., 2023). Furthermore, elevated cytosolic Ca^{2+} promotes rapid mitochondrial calcium uptake, which stimulates oxidative phosphorylation to enhance ATP generation while simultaneously increasing mitochondrial ROS (mtROS) production as a metabolic byproduct (Xu et al., 2022). Membrane potential dynamics, mitochondrial calcium flux, and mtROS generation were examined to assess the effect of MRC2-mediated PTL endocytosis on mitochondrial function in RAW 264.7 macrophages. The data demonstrated that endocytosed PTL co-localized with

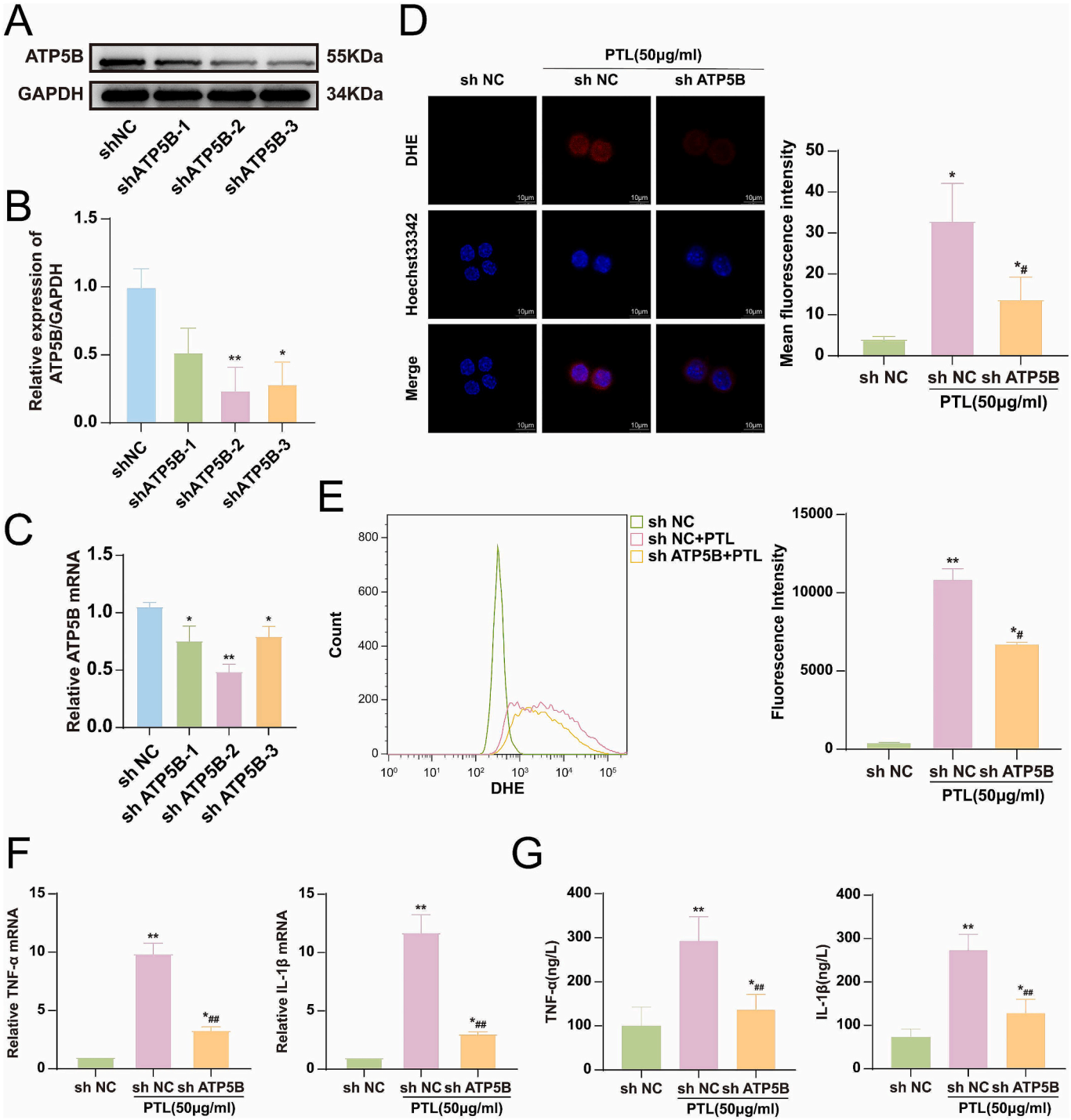


Fig. 7. Knockdown of ATP5B in RAW264.7 cells reduced (i) PTL-induced ROS release and (ii) the expression levels of TNF-α and IL-1β. (A and B) Western blot analysis and (C) RT-qPCR verified the transfection efficiency. $^{*}p < 0.05$, $^{**}p < 0.001$ compared with the shNC group. The effect of ATP5B knockdown on ROS release was assessed by (D) laser scanning confocal microscopy and (E) flow cytometry, with nuclei counterstained with Hoechst 33342 (blue). Scale bar: 10 μm. The right panels show quantitative analysis of ROS fluorescence intensity. The effects of ATP5B knockdown on the release levels of TNF-α and IL-1β were determined by (F) RT-qPCR and (G) ELISA. $^{*}p < 0.05$, $^{**}p < 0.001$ compared with the shNC group; $^{\#}p < 0.05$, $^{\#\#}p < 0.001$ compared with the PTL-treated shATP5B group. Data are presented as the mean ± SD (n = 6 for ELISA; n = 3 for Western blotting, RT-qPCR, laser scanning confocal microscopy, and flow cytometry). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

mitochondria and induced a characteristic pathological triad: marked ΔΨm dissipation, elevated mitochondrial calcium accumulation, and increased mtROS generation. Collectively, these results demonstrated that PTL induced mitochondrial injury through MRC2-mediated endocytosis in RAW 264.7 macrophages.

To further elucidate the mechanism underlying PTL-induced mitochondrial injury, Co-IP-MS/MS was employed to identify potential mitochondrial interacting proteins. Among the identified mitochondrial-associated proteins, ATP5B exhibited the highest enrichment intensity. ATP5B is localized to the inner mitochondrial membrane and functions as a critical catalytic subunit of the F1 sector of mitochondrial ATP synthase, where it mediates the phosphorylation

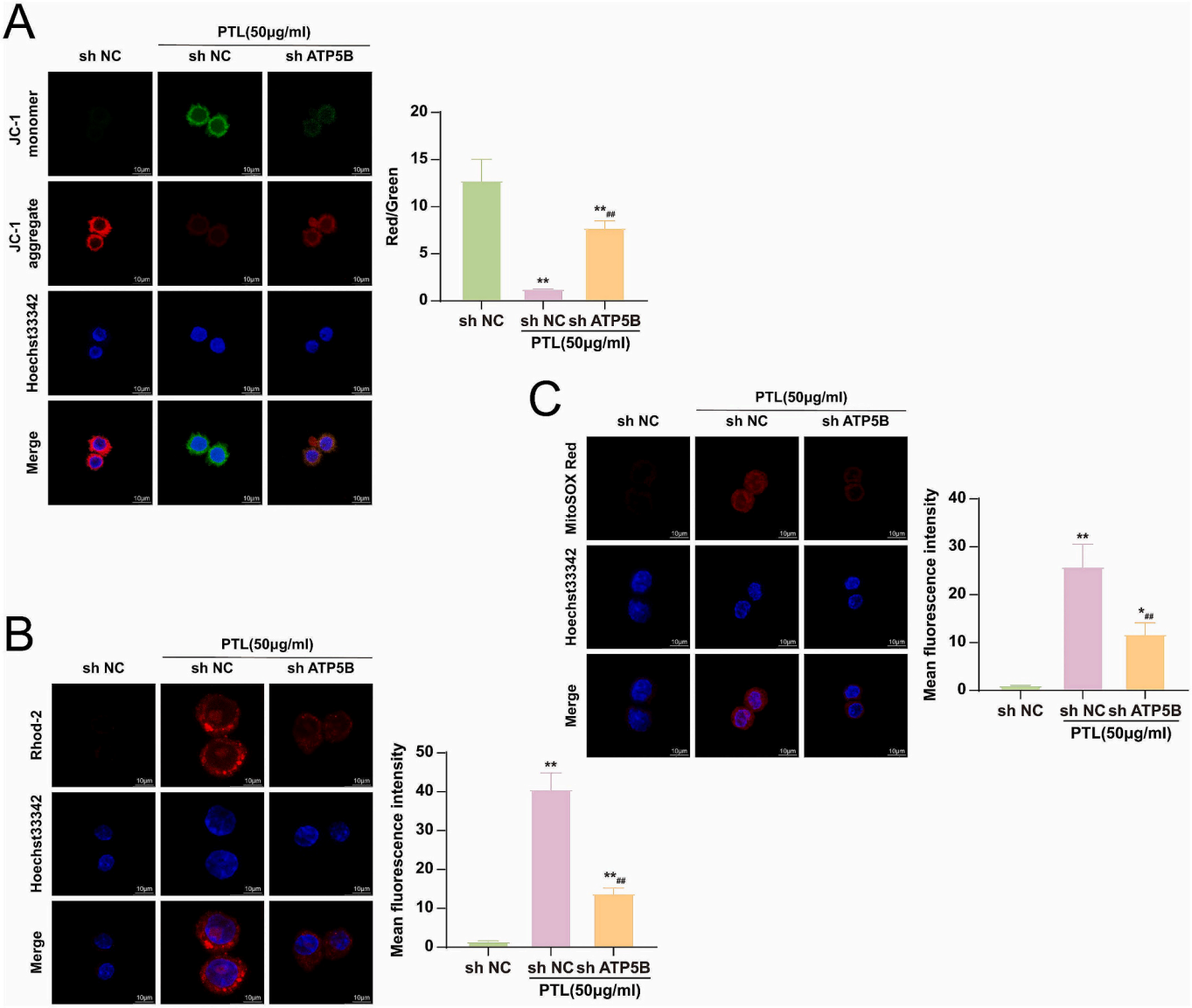


Fig. 8. Knockdown of ATP5B elevated the mitochondrial membrane potential while reducing mitochondrial calcium and mitochondrial ROS levels in PTL-treated RAW264.7 cells. (A) Mitochondrial membrane potential was assessed using laser scanning confocal microscopy. (B) Changes in mitochondrial calcium levels were visualized by laser scanning confocal laser scanning microscopy. (C) Mitochondrial reactive oxygen species (ROS) were detected via laser scanning confocal microscopy. Nuclei were counterstained with Hoechst 33342 (blue). Scale bar, 10 μm. The quantitative analysis results were shown in the right panel. $^{*}p < 0.05$, $^{**}p < 0.001$ compared with the shNC group; $^{\#\#}p < 0.001$ compared with the PTL-treated shNC group. Data are presented as the mean ± SD. All experiments included n = 3 independent biological replicates per group, with technical triplicates in each. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

step in ATP biosynthesis (Chen et al., 2018). To validate the interaction between PTL and ATP5B, molecular docking analysis was first performed, which revealed an interaction area of 1957.7 Å^2 and a binding free energy of −3.1 kcal/mol, supporting the structural feasibility of their interaction. Meanwhile, laser scanning confocal microscopy showed clear co-localization of PTL with ATP5B in cells, further corroborating their association at the cellular level. To quantitatively determine the direct binding affinity between PTL and ATP5B, microscale thermophoresis (MST) binding assays were further conducted. The results demonstrated that the normalized fluorescence signals exhibited a typical sigmoidal dose-dependent change with increasing PTL concentrations, and fitting the data to a sigmoidal binding model yielded a dissociation constant (Kd) of 3.52 ± 0.31 μM, confirming a direct binding affinity between PTL and ATP5B. Collectively, the structural predictions from molecular docking, the cellular co-localization evidence, and the quantitative MST data collectively confirmed the direct interaction between PTL and ATP5B at multiple levels.

Beyond its canonical bioenergetic function, ATP5B also participates in mitochondrial stress signaling, ROS regulation, and inflammatory pathway activation (Chung et al., 2020; Wang et al., 2021a). Perturbation of ATP5B function induces electron transport chain dysfunction, electron leakage, oxidative stress amplification, and NLRP3 inflammasome activation, ultimately contributing to mitochondrial dysfunction (Chung et al., 2020; Huang et al., 2016; Pei et al., 2024). Combined with the validation results from molecular docking, co-localization, and MST assays in the present study, the binding of PTL to ATP5B may interfere with the normal function of ATP5B, thereby serving as a critical molecular event underlying PTL-induced mitochondrial injury.

Previous studies have demonstrated that ATP5B participates in mitochondrial stress and inflammatory regulation. In perfluorooctane sulfonate-exposed hepatocytes, ATP5B promoted mPTP opening and NLRP3 inflammasome activation through regulation of VDAC1

oligomerization, thereby contributing to inflammatory liver injury (Ma, 2024). ATP5B knockdown alleviated cigarette smoke extract-induced apoptosis and reduced inflammatory cytokine production (Zhang et al., 2025). Moreover, several plant lectins have been reported to induce ROS accumulation and increase TNF-α and IL-1β production in RAW264.7 macrophages, thereby triggering inflammatory responses (Pan et al., 2015). Given prior evidence demonstrating the involvement of ATP5B in cellular stress and inflammatory signaling, together with the established notion that the release of inflammatory cytokines and cellular oxidative stress contribute to mitochondrial dysfunction (Ueda and Suwanmanee, 2022). these findings suggested that ATP5B may serve as a key mediator of PTL-induced mitochondrial damage and inflammatory activation.

To further investigate whether PTL induced inflammatory responses and mitochondrial damage through direct interaction with ATP5B, lentivirus-mediated transfection was used to knock down ATP5B in RAW264.7 macrophages. Among the shRNA constructs tested, shATP5B-2 exhibited the highest knockdown efficiency and was therefore selected for subsequent experiments. Functional assays demonstrated that ATP5B knockdown significantly attenuated PTL-induced ROS generation, as assessed by both laser scanning confocal microscopy and flow cytometry. Furthermore, PTL-induced upregulation of TNF-α and IL-1β at both mRNA and protein levels was markedly suppressed following ATP5B knockdown. In terms of mitochondrial function, PTL treatment reduced mitochondrial membrane potential, as indicated by a decreased red/green fluorescence ratio, while simultaneously promoting mitochondrial calcium accumulation and mitochondrial ROS production. Notably, ATP5B knockdown significantly alleviated these PTL-induced mitochondrial alterations, including mitochondrial membrane potential dissipation and excessive mitochondrial calcium and ROS release. Knockdown of ATP5B exerted a protective effect by reducing the interacting sites available for PTL, thereby attenuating PTL-triggered inflammatory responses and mitochondrial damage signaling. Collectively, these findings demonstrated that PTL induced mitochondrial damage and inflammatory responses through its interaction with ATP5B in RAW 264.7 macrophages.

Beyond its pro-inflammatory effects, PTL, like many other plant lectins, has attracted attention for its potential anti-tumor activities. Plant lectins have attracted increasing attention because of their dual biological properties, including inflammatory toxicity and anti-tumor activity (Shi et al., 2016). Representative lectins, such as *Polygonatum odoratum* lectin (POL), *mistletoe* lectin (ML), and ConA, exert pronounced anti-tumor effects by modulating cancer-associated signaling pathways, including Bcl-2, caspase, and PI3K/Akt, thereby inducing apoptosis and autophagy (Jiang et al., 2015). Current research has indicated that ATP5B mRNA expression varies substantially among cancer types, suggesting context-dependent regulation and functional heterogeneity (Gale et al., 2020). Specifically, ATP5B expression is markedly downregulated in colorectal cancer but upregulated in lung, cervical, and cholangiocarcinomas (Wang et al., 2021a, 2021b). Accumulating evidence supports the potential utility of ATP5B as prognostic biomarker in cancer (Liu et al., 2021).

The present study primarily elucidated the molecular mechanism through which PTL promoted inflammatory responses via MRC2-mediated endocytosis in RAW 264.7 macrophages. These findings may provide a basis for further investigations into the immunomodulatory properties of PTL. However, the practical therapeutic application of plant lectins, including PTL, faces several challenges, including limited oral bioavailability due to potential proteolytic degradation in the gastrointestinal tract, the need for appropriate administration routes to achieve effective tumor targeting, and potential immunogenicity concerns. Nevertheless, accumulating evidence has demonstrated that PTL exhibits significant antiproliferative activity against various tumor cell lines, including Sarcoma 180, HeLa, K562 (Zuo et al., 2012), and Bel-7404 hepatocellular carcinoma cells, through induction of apoptosis (Zhou et al., 2014). Moreover, plant lectins have been shown to suppress cancer cells by modulating various signaling pathways, including the caspase family, mitochondrial-mediated ROS-p38-p53 pathway, PI3K/Akt, and Wnt/β-catenin (Bhutia et al., 2019; Jiang et al., 2015). Notably, mistletoe lectins have already been developed and clinically applied for cancer treatment (Beuth, 1997), suggesting that plant lectins hold promise as therapeutic agents. Given the ability of certain plant lectins to recognize specific carbohydrate moieties on tumor cells and modulate cancer-associated signaling pathways, PTL may have potential as a candidate for further anti-cancer research, provided that these challenges are adequately addressed.

5. Conclusion

The present study establishes an integrated pathway from *Pinellia ternata* lectin (PTL) endocytosis to mitochondrial targeting and subsequent pro-inflammatory responses. We demonstrate that PTL enters RAW 264.7 cells via mannose receptor C-type 2 (MRC2)-mediated endocytosis and directly interacts with ATP5B, inducing mitochondrial damage, oxidative stress, and ultimately inflammatory cytokine release. These findings elucidate the toxicological mechanism of *Pinellia ternata* lectin. Furthermore, this study reveals that exogenous lectins disrupt innate immune homeostasis through organelle-specific injury, expanding our understanding of lectin immunotoxicity.

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CRediT authorship contribution statement

Yangyi Huang: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Jinfei Li:** Writing – review & editing, Validation, Methodology, Investigation, Data curation, Conceptualization. **Yue Lu:** Validation, Methodology, Investigation. **Jie Cao:** Investigation. **Xinzhi Wang:** Resources. **Hao Wu:** Supervision, Project administration, Funding acquisition. **Min Shen:** Validation, Resources. **Hongli Yu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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